

A Baseline for ViCo Conversational Head Generation Challenge

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ABSTRACT

ViCo conversational head generation challenge is organized in conjunction with ACM Multimedia 2022. The challenge includes two tasks: 1) generating a vivid talking head according to a reference image and driving audio, and 2) synthesizing a listening head conditioned on the reference image and speaker's behaviors. For talking head generation, we design a three-stage method to generate high-quality and lip-sync talking head videos with natural head poses, which wins 3rd place and People's Selection Award. On the listening head generation track, we adopt MakeltTalk to generate vivid head movements of listeners, which achieves 2nd place in the competition.

CCS CONCEPTS

• Information systems → Multimedia content creation.

KEYWORDS

Talking head generation, Listener generation

ACM Reference Format:

Meng Liu, Shuyan Zhai, Yongqiang Li, Weili Guan, and Liqiang Nie. 2022. A Baseline for ViCo Conversational Head Generation Challenge. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3503161.3551569>

1 INTRODUCTION

ACM Multimedia 2022 Conversational Head Generation Challenge includes two tracks, Vivid Talking Head Video Generation and Responsive Listening Head Video Generation. As illustrated in Figure 1, the former generates talking head video conditioned on the image and the audio information of the specific speaker, and the latter generates listening head video conditioned on the image of the listener and the speaker's behaviors in real-time. Besides, the organizers published a dataset, named ViCo. During the competition, participants are restricted to train their algorithms on the released training set, while additional data of the target identities

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MM '22, October 10–14, 2022, Lisboa, Portugal

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ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551569>

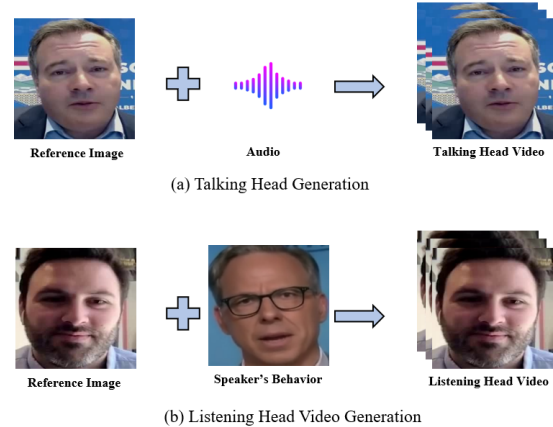


Figure 1: Illustrations of Vivid Talking Head Generation and Responsive Listening Head Video Generation.

is unusable. Moreover, public pre-trained models and additional unlabeled data for pre-training are allowed in the competition.

We rank 2nd on Responsive Listening Head Video Generation track and 3rd on Vivid Talking Head Video Generation track¹. Meanwhile, we obtain the People's Selection award² on Vivid Talking Head Video Generation.

2 TALKING HEAD GENERATION

As shown in Figure 2, our method consists of three components: head pose generator, lip synchronization retainer, and video quality optimizer. Therein, head pose generator is designed to synthesize videos with natural head movements. Lip synchronization retainer aims to refine the lip movements of the videos generated by the first stage, ensuring synchronize with the input audio. As to video quality optimizer, it could preserve details of the talking person, therefore improving the image quality of the generated video. In the following, we will elaborate each component sequentially.

2.1 Head Pose Generator

Given the audio, we first split it into segments in sync with the video, i.e., 30 frames per second. And then for each audio segment, we extract the acoustic feature, including Mel Frequency Cepstrum Coefficients (MFCC) with the corresponding MFCC Delta and Delta-Delta feature, the energy, loudness, and zero-crossing rate (ZCR). In

¹The final ranking is based on the number of first place across all metrics.

²Ten experts from production, user experience development and research areas select one team for each track's People's Selection Award.

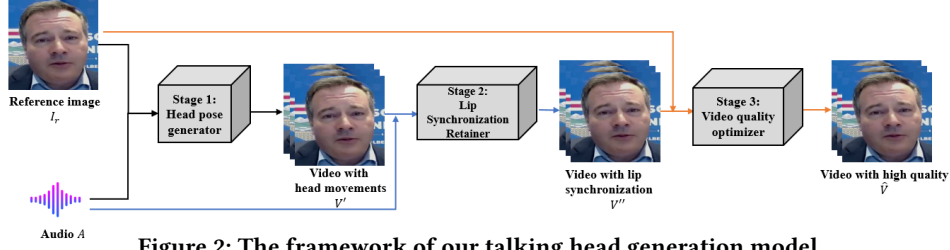


Figure 2: The framework of our talking head generation model.

this paper, we denote the acoustic feature sequence by $a_{1:T}$, where $a_i \in \mathbb{R}^{45}$ refers to the acoustic feature of the i -th frame.

For the reference image, we utilize 3D face reconstruction model [5] to obtain 3DMM [2] coefficients, including identity representation $\alpha \in \mathbb{R}^{80}$, expression representation $\beta \in \mathbb{R}^{64}$, textual representation $\delta \in \mathbb{R}^{27}$, head pose representation $\rho \in \mathbb{R}^6$, and lighting parameters $\gamma \in \mathbb{R}^{27}$. To synthesize the video of the reference person, the identity parameter is constant. Meanwhile, for convenience of subsequent rendering, we also fix the lighting and textual coefficients during training.

To generate talking heads with natural head movements and expression, we develop an encoder-decoder framework. To be specific, we first employ layer-norm and a fully connected layer to get head pose embedding P_0 and expression embedding E_0 , respectively. Meanwhile, given the audio feature sequence $a_{1:T}$, we employ the audio encoder E_A , which is composed of a four-layer bidirectional LSTM and a fully connected layer, obtaining audio latent feature sequence $S_{1:T}$. Afterwards, we concatenate the audio embedding S_1 with the head pose embedding P_0 as the initial vector, and put it into a head pose decoder D_P , which is composed of a two-layer LSTM and a fully connected layer. Finally, we could generate natural head motion sequence $P_{1:T}$.

Similarly, we adopt the same architecture to predict expression coefficients. Note that during decoding sequential expression information, we concatenate S_{i-1} , S_i , and S_{i+1} as the audio vector rather than S_i at each step. Having obtained the predicted pose and expression coefficients, we employ the pre-trained rendering model [10] to get talking faces with head movements V' .

2.2 Lip Synchronization Retainer

Although V' with vivid head poses, it suffers from significant lip out-of-sync problem. As Wav2lip [9] excels at producing accurate lip movements of an arbitrary identity, we adopt Wav2lip pre-trained on LRS2 [1] as our lip synchronization retainer. Note that it only changes the lower face of the talking person, especially the mouth. Therefore, we obtain talking faces with both vivid head poses and accurate lip movements, denoted by V'' .

2.3 Video Quality Optimizer

As the model limitation of Wav2lip, the resolution of V'' is relatively low. To improve the quality of V'' , we adopt the First-Order-Motion model [11] pre-trained on VoxCeleb [8] as our video quality optimizer, to generate high-quality talking head video $\hat{V}$. It could animate the person of an input image according to the motion of a driving video, and maintain the clarity of the input image.

3 LISTENING HEAD GENERATION

Regarding listening head generation task, we utilize MakeItTalk [14] to generate vivid listening head with natural head movements and facial expressions. It is a two-stage facial video generation model with 3D facial landmarks as an intermediate product. Note that we directly utilize the model parameters published by the authors, which is pre-trained on Obama Weekly Address [12] and Voxceleb2 [4].

4 EXPERIMENT

4.1 Evaluation Metrics

We utilize the evaluation metric provided by the challenge to justify the effectiveness of our model.

- Generation quality (image level): SSIM [13], CPBD [3], PSNR;
- Generation quality (feature level): FID [7];
- Identity preserving: Cosine Similarity (Arcface);
- Expression: L1 distance of 3dmm exp features;
- Head motion: L1 distance of 3dmm angle & trans features;
- Lip sync: AV offset and AV confidence (SyncNet);
- Lip landmark distance: L1 distance of lip landmarks.

4.2 Implementation Details

For head pose generator, we set teacher forcing ratio [6] as 1 at the beginning (decay 0.05 per epoch). We use the Adam optimizer with an initial learning rate of 2×10^{-3} (decayed exponentially by 0.5 every 10 epochs) as well as $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We set default batch size as 256. During pre-training the lip sync retainer, we set the batch size to 80. We use the Adam optimizer with an initial learning rate of 10^{-4} as well as $\beta_1 = 0.5$ and $\beta_2 = 0.999$ for both the generator and discriminator. We fine-tuned the model on the training set of ViCo dataset with the same setup. During pre-training the video quality optimizer, the batch size is set to 36. We use the Adam optimizer with an initial learning rate of $2e^{-4}$ as well as $\beta_1 = 0.5$ and $\beta_2 = 0.999$. We fine-tuned the model on training set of ViCo dataset with the same setup.

4.3 Performance Comparison

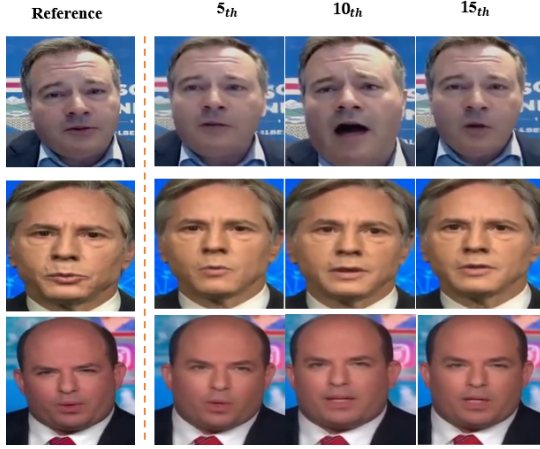
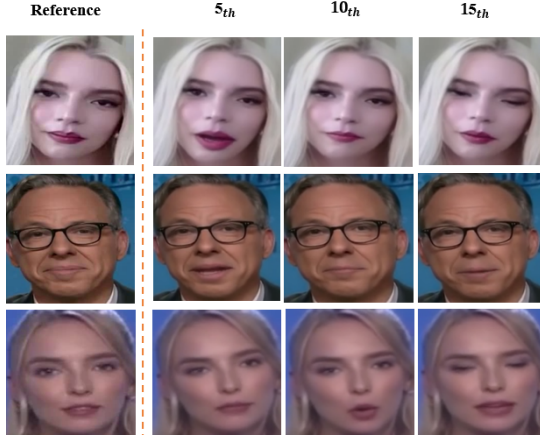
The experimental results on listening head video generation are reported in Table 1. We can see that our model achieve better performance in term of SSIM, CPBD, and FID. This reflects that our model is good at ensuring image quality. Table 2 summarizes the performance comparison on the talking head generation track. As can be seen from the results, although our image level and feature level metrics are slightly lower than the other two teams, we have

Table 1: Performance Comparison on listening head generation.

Metrics	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	ExpFD↓	AngleFD↓	TransFD↓
1 st Team	0.580	0.111	18.512	21.350	0.247	0.116	0.034	0.054
Ours	0.582	0.088	18.491	26.675	0.247	0.133	0.038	0.073

Table 2: Performance comparison on talking head generation.

Metrics	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	ExpFD↓	AngleFD↓	TransFD↓	AVOffset↓	AVConf↑	LipLMD↓
1 st Team	0.608	0.175	17.767	29.709	0.144	0.131	0.056	0.057	2.167	0.867	10.101
2 nd Team	0.574	0.218	17.179	24.678	0.124	0.137	0.056	0.056	1.200	3.153	10.646
Ours	0.565	0.134	16.546	25.050	0.135	0.137	0.063	0.064	1.000	8.045	10.900

**Figure 3: Talking head generation results of our model.****Figure 4: Listening head generation results of our model.**

a huge advantage in audio and video synchronization. We get first in AVConf metric.

4.4 Qualitative Results

In Figure 3, we display some results on talking head generation. It can be seen that our generated videos have natural head movement and high resolution of the mouth. Figure 4 shows some results on listening head generation. We can see that our video is relatively clear and the head movement is smooth.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China under Grant 62006142 and Grant U1936203, in part by the Shandong Provincial Natural Science Foundation for Distinguished Young Scholars under Grant ZR2021JQ26, in part by the Major Basic Research Project of Natural Science Foundation of Shandong Province under Grant ZR2021ZD15, in part by the Science and Technology Innovation Program for Distinguished Young Scholars of Shandong Province Higher Education Institutions under Grant 2021KJ036.

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